

WASP-50b

R-0726

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-50_231223 · created 2026-07-28T00:23:50+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460301.644 +/- 0.098 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.175 +/- 0.05
Transit depth (Rp/Rs)^2	3.06 +/- 1.74 [%]
Orbital Inclination (inc)	87.99 +/- 2.89
Airmass coefficient 1 (a1)	6.74 +/- 4.13
Airmass coefficient 2 (a2)	-1.22 +/- 1.11
Scatter in the residuals of the lightcurve fit is	68.19 %
Best Comparison Star	#2 - [622, 99]
Optimal Aperture	2.41
Optimal Annulus	3.0
Transit Duration (day)	0.047 +/- 0.037

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.175 ± 0.05 vs 0.1404 (deviation 0.58σ)
Duration fit vs published	0.047 d vs 0.0754 d (deviation 0.64σ)
Mid-time O–C	-34.6 min
Depth SNR	2.9
Residual scatter	68.19 %

Light curve

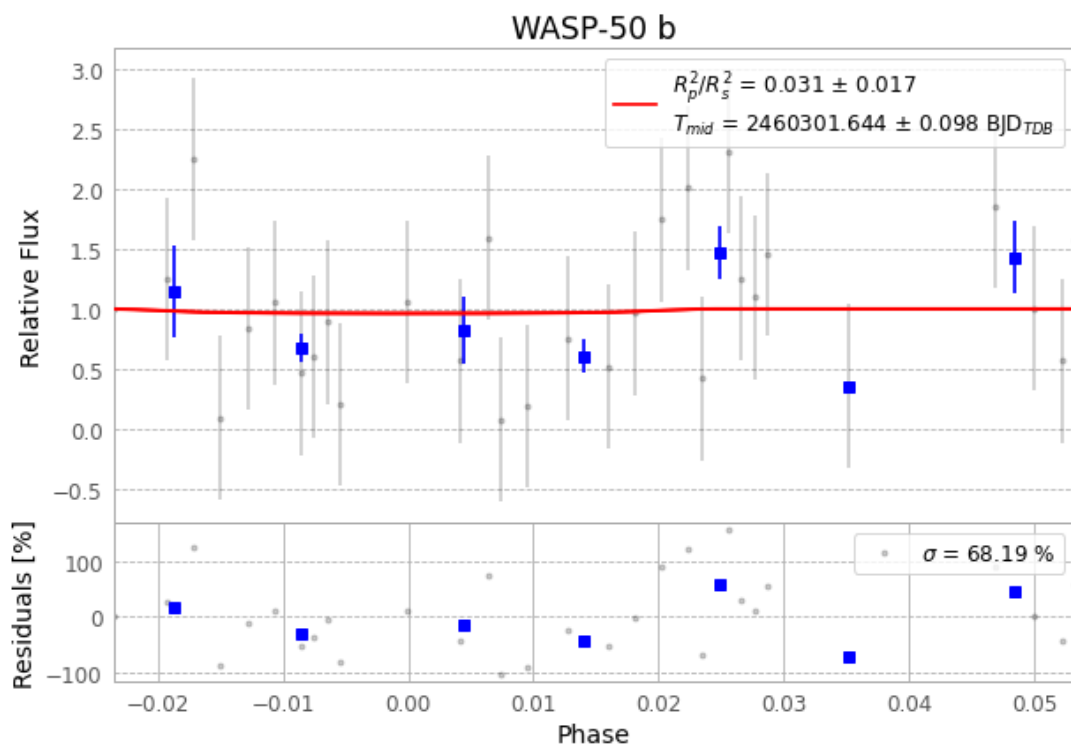


Figure 1 — FinalLightCurve_WASP-50 b_2023-12-22.png (SHA-256 438e399845bf946e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [477, 148], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 08c1830bfa5179cf...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

72 FITS frames (47 MB), WASP-50231223021512.FITS → WASP-50231223055115.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-50-231223.log (480b8409fa3f...), FinalLightCurve_WASP-50 b_2023-12-22.png (438e399845bf...), FinalLightCurve_WASP-50 b_2023-12-22.csv (c8b03d6951d8...)

Compute resources

Wall time 110.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 00:23Z.